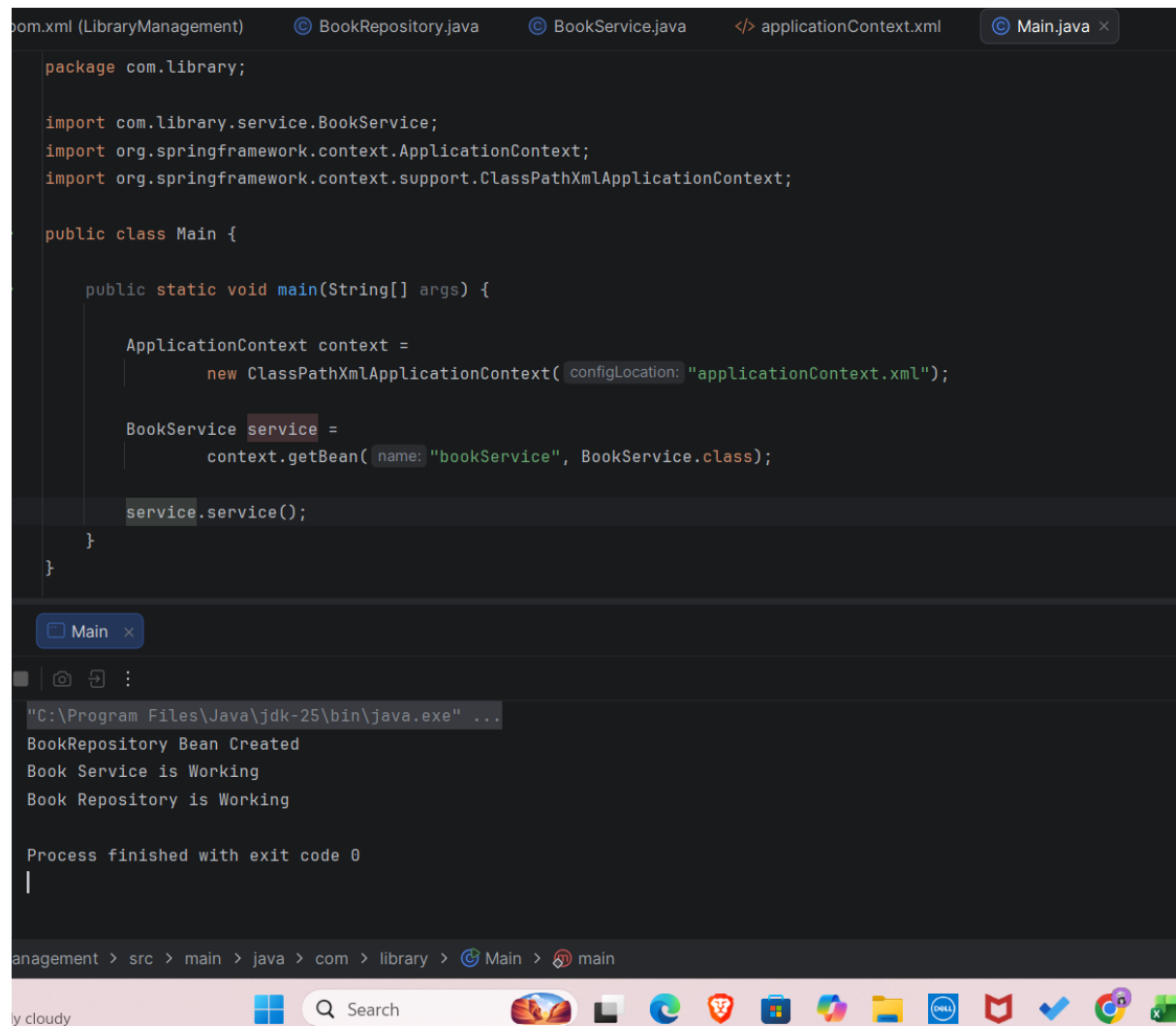


Spring Core and Maven Mandatory Hands-on

Spring Core_Maven

Exercise 1: Configuring a Basic Spring Application



The screenshot displays an IDE with the following components:

- Editors:** pom.xml (LibraryManagement), BookRepository.java, BookService.java, applicationContext.xml, and Main.java.
- Main.java Code:**

```
package com.library;

import com.library.service.BookService;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;

public class Main {

    public static void main(String[] args) {

        ApplicationContext context =
            new ClassPathXmlApplicationContext( configLocation: "applicationContext.xml");

        BookService service =
            context.getBean( name: "bookService", BookService.class);

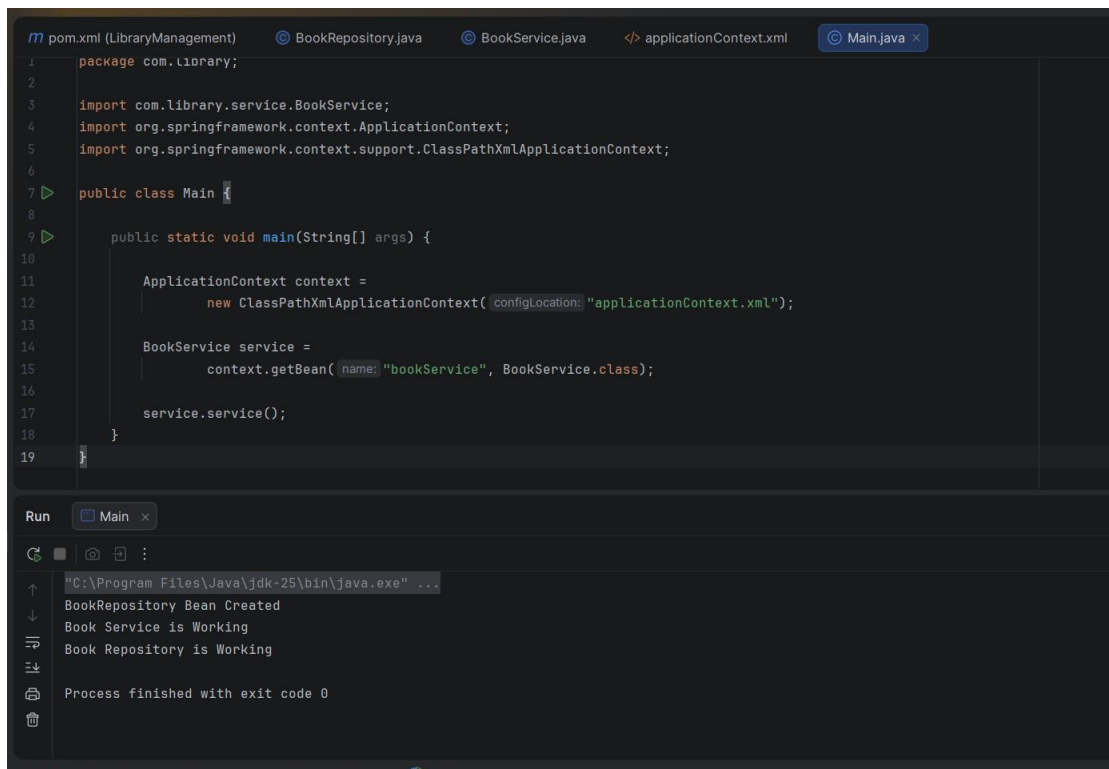
        service.service();
    }
}
```
- Run Console:**

```

Main
"C:\Program Files\Java\jdk-25\bin\java.exe" ...
BookRepository Bean Created
Book Service is Working
Book Repository is Working

Process finished with exit code 0
|
```
- Bottom Bar:** Shows the file path: management > src > main > java > com > library > Main > main.
- Taskbar:** Includes the Windows Start button, a search bar, and various application icons.

Exercise 2: Implementing Dependency Injection



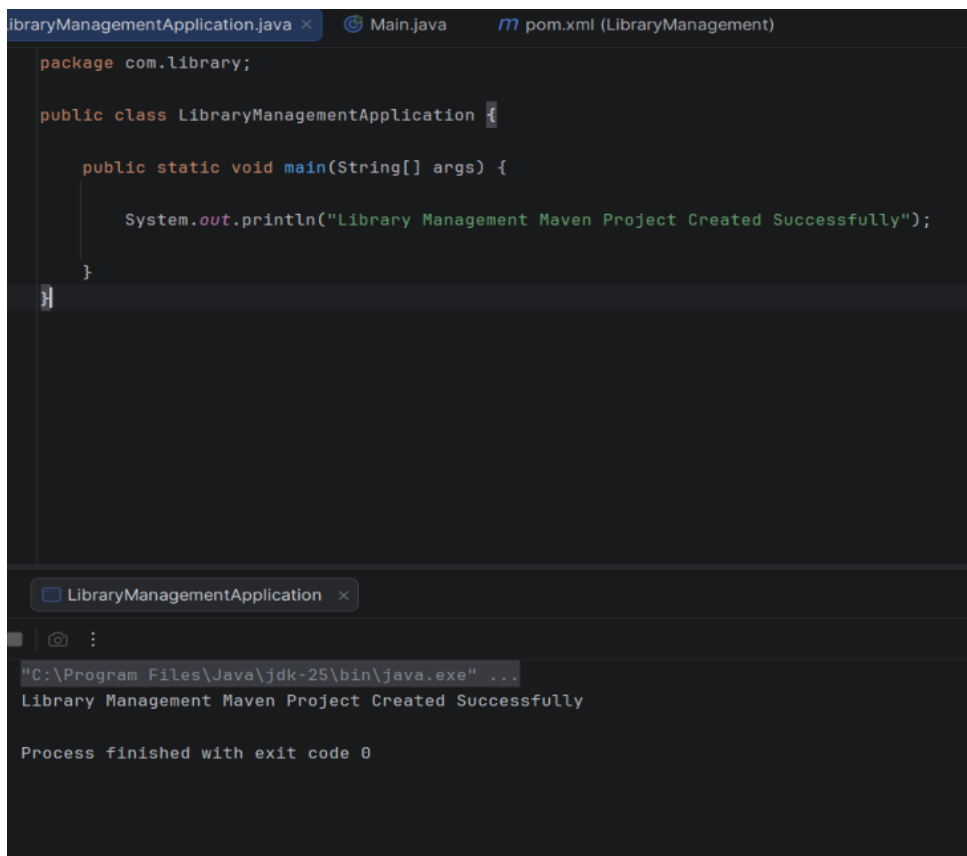
The screenshot shows an IDE with a project named 'LibraryManagement'. The 'Main.java' file is open, showing the following code:

```
1 package com.library;
2
3 import com.library.service.BookService;
4 import org.springframework.context.ApplicationContext;
5 import org.springframework.context.support.ClassPathXmlApplicationContext;
6
7 public class Main {
8
9     public static void main(String[] args) {
10
11         ApplicationContext context =
12             new ClassPathXmlApplicationContext("applicationContext.xml");
13
14         BookService service =
15             context.getBean("bookService", BookService.class);
16
17         service.service();
18     }
19 }
```

The 'Run' console shows the following output:

```
Run Main x
"C:\Program Files\Java\jdk-25\bin\java.exe" ...
BookRepository Bean Created
Book Service is Working
Book Repository is Working
Process finished with exit code 0
```

Exercise 4: Creating and Configuring a Maven Project



The screenshot shows an IDE with a project named 'LibraryManagement'. The 'LibraryManagementApplication.java' file is open, showing the following code:

```
package com.library;

public class LibraryManagementApplication {

    public static void main(String[] args) {

        System.out.println("Library Management Maven Project Created Successfully");

    }

}
```

The 'Run' console shows the following output:

```
LibraryManagementApplication x
"C:\Program Files\Java\jdk-25\bin\java.exe" ...
Library Management Maven Project Created Successfully
Process finished with exit code 0
```

Exercise 5: Configuring the Spring IoC Container

```
1 package com.library;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 import com.library.service.BookService;
7
8 public class LibraryManagementApplication {
9
10     public static void main(String[] args) {
11
12         ApplicationContext context =
13             new ClassPathXmlApplicationContext("applicationContext.xml");
14
15         BookService service = context.getBean("bookService", BookService.class);
16
17         service.addBook();
18     }
19 }
```

Run Main

```
"C:\Program Files\Java\jdk-25\bin\java.exe" ...
Book saved successfully
Book added through BookService
Process finished with exit code 0
```

Exercise 7: Implementing Constructor and Setter Injection

```
1 package com.library;
2
3 import org.springframework.context.ApplicationContext;
4 import org.springframework.context.support.ClassPathXmlApplicationContext;
5
6 public class Main {
7
8     public static void main(String[] args) {
9
10         ApplicationContext context =
11             new ClassPathXmlApplicationContext("applicationContext.xml");
12
13         BookService service = context.getBean("bookService", BookService.class);
14
15         service.showBook();
16     }
17 }
```

Run Main

```
"C:\Program Files\Java\jdk-25\bin\java.exe" "-javaagent:C:\Users\BHASHINI P\AppData\Local\Programs\IntelliJ IDEA 2026.1.4\lib\idea_rt.jar=55995" -Dfile.encoding=UTF-8 -Dsun.st
Constructor Injection Executed
Setter Injection Executed
BookRepository: Displaying Book Information
Process finished with exit code 0
```

